higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

9 APRIL 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1	1.1.1	C		
	1.1.2	B		
	1.1.3	A		
	1.1.4	A		
	1.1.5	C		
			(5 × 1)	(5)
1.2	1.2.1	C		
	1.2.2	A		
	1.2.3	E		
	1.2.4	B		
	1.2.5	F		
			(5 × 1)	(5)
1.3	1.3.1	True		
	1.3.2	True		
	1.3.3	False		
	1.3.4	True		
	1.3.5	False		
			(5 × 1)	(5)
1.4	1.4.1	2421 2421 2421✓ 1011 1100 0100✓ =10111100,0100✓		
	1.4.2	8421 8421 8421✓ 0101 0110 0100✓ =1010110,0100✓		
	1.4.3	5 6 4✓ +3 +3 +3 8 9 7✓ =10001001,0111✓		
			(3 × 3)	(9)
				[24]

QUESTION 2

2.1 $19,75_{10}$ = 010011, 110 = 101100, 001✓

+ 1

101100, 010✓

$15,125_{10}$ = 001111, 001✓

+ 1

110000, 111✓

101100, 010
110000, 111
011101, 001✓
= $35,1_8$✓

2.2 F, C_{16} = 1111, 1100✓

5, 4_8 = 101, 100✓

101,101✓
1011) 111111000
-1011
10011
-1011✓
10000
-1011✓
10100
-1011
1001
= 9_{10}✓

2.3

111100111✓
1101
111100111✓
000000000
111100111✓
111100111✓
11000101, 11011✓

= $305,66_8$✓

2.4 1100, 010 = 0011, 101✓

1011, 001✓
0011, 101
1110, 110✓
0001, 001✓

= -0001, 001✓
= $-1,1_8$✓

(4 × 6) **[24]**

QUESTION 3

3.1 $F = \overline{A\,\overline{C}\,(B + \overline{C})\,\overline{B}}$

$= \overline{A\,\overline{C}}\,(B\,\overline{B} + \overline{B}\,\overline{C})$ ✓✓

$= \overline{A} + C\,(0 + \overline{B}\,\overline{C})$ ✓

$= \overline{A}\,\overline{B}\,\overline{C} + B\,\overline{C}\,C$✓✓

$= \overline{A}\,\overline{B}\,\overline{C}$✓

3.2 LET AB = K

= K B✓

$= K\,\overline{B} + \overline{K}\,B$✓

$= A\,B\,\overline{B} + \overline{AB}B$✓

$= (\overline{A} + \overline{B})\,B$✓

$= \overline{A}\,B + \overline{B}\,B$✓

$= \overline{A}\,B$✓

(2 × 6) **[12]**

QUESTION 4

4.1 4.1.1 The output of one flip-flop is used to activate the next flip-flop being fed directly onto this clock input.

4.1.2 Synchronous counters are simultaneously clocked by the same clock input pulses.

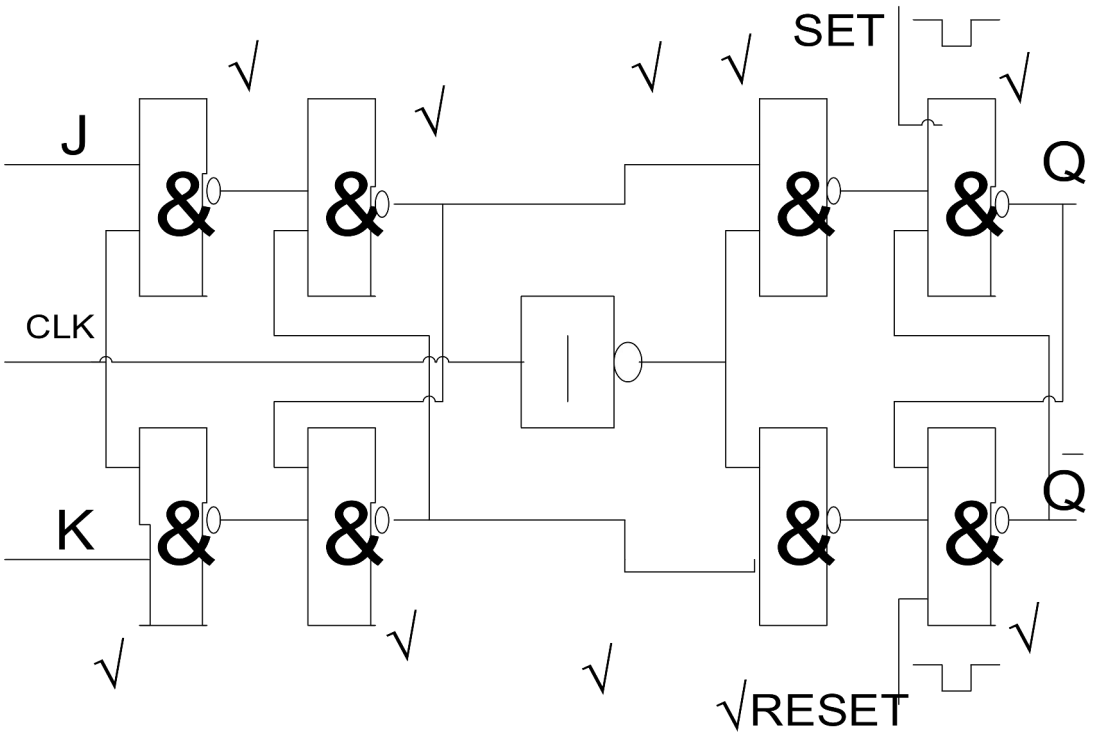
(2 × 2) (4)

4.2

Position number	Master	Slave
1	4.2.1 closed✓	open
2	closed	4.2.2 closed✓
3	open	4.2.3 closed✓
4	4.2.4 closed✓	closed
5	4.2.5 closed✓	open
6	closed	4.2.6 open✓

(6)

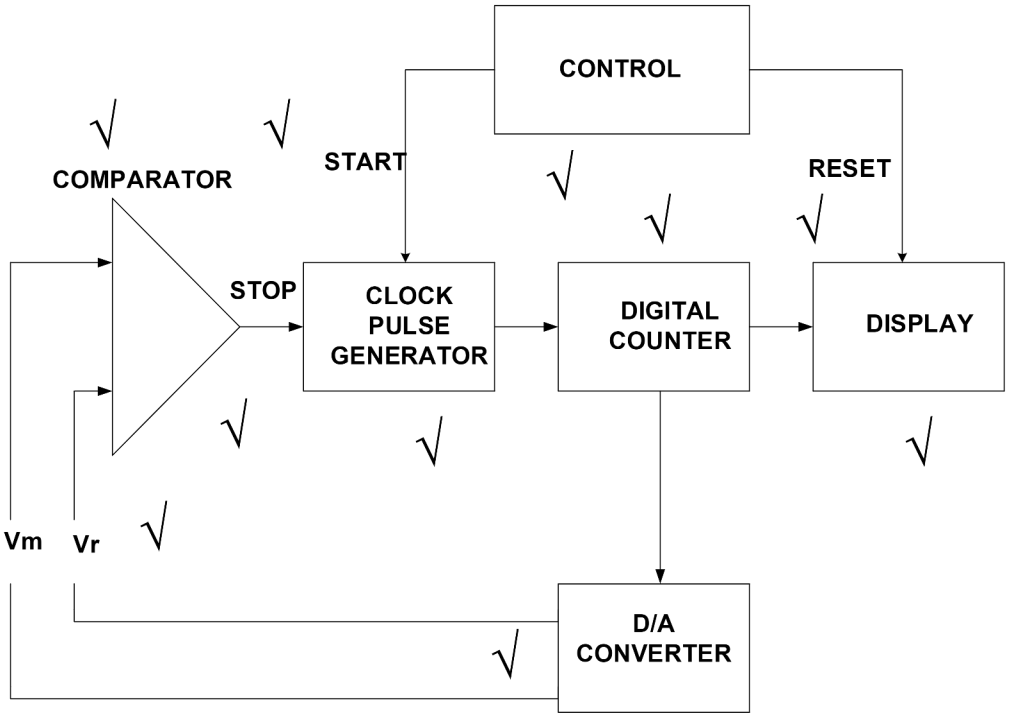
4.3



(10)
[20]

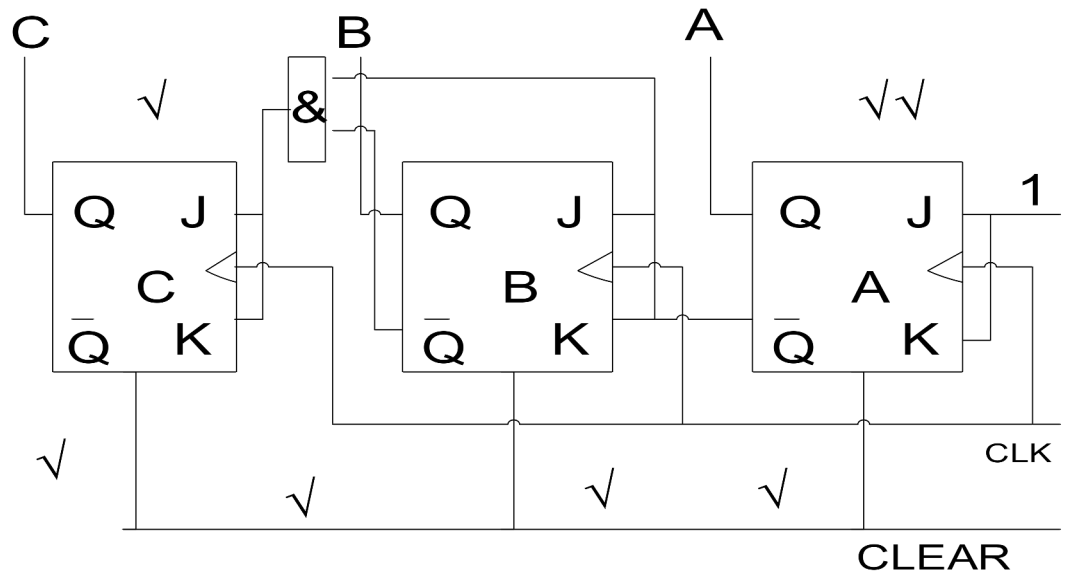
QUESTION 5

5.1



(10)

5.2



✓✓✓

CLOCK	C	B	A
0	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	0	1	1
6	0	1	0
7	0	0	1
8	0	0	0

(10)
[20]

TOTAL: 100

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